A THREE-PART SERIES · MAY 2026

The GUI Agents Trilogy

MolmoWeb's training inversion, the open-release reproducibility gap, and the verifiers that overcount success — three layers of the same story.

PART 1

The Training Inversion

PART 2

Inside the Harness

PART 3

The Verification Gap

SERIES MAP

Three posts, one through-line

The evaluation stack has systematic errors at every layer — the training signal, the implementation, the measurement — and they compound in one direction: toward overstated capability.

PART 1

The Training Inversion

Synthetic AxTree trajectories beat human demonstrations by 17 points on WebVoyager at matched scale.

PART 2

Inside the Harness

The repo went complete — and complete reveals defaults, dataset dual-use, and a judge loop.

PART 3

The Verification Gap

Canonical verifiers overcount success — a 60% false-positive rate on one benchmark.

01

PART 1 · THE TRAINING INVERSION · MAY 4, 2026

The Training Data Inversion

MolmoWeb is the most open visual web agent to date — but its most durable finding is an ablation, not a benchmark number: synthetic trajectories beat human demonstrations.

PART 1 · THE HOOK

28 steps. Wrong product.

The task: "Find the pricing of the latest Apple AirTag." Two ways to solve it, one inside the training data — and they don't look alike.

HUMAN ANNOTATOR

28 steps

- Scrolled up then down 14 times
- Lost the task category entirely
- Navigated genuine uncertainty

Submitted: "The latest iPhone model was 17 pro max..."

SYNTHETIC TEACHER (AXTREE)

11 steps

- No hover detours
- No up-then-down scrolling
- Reads element semantics directly

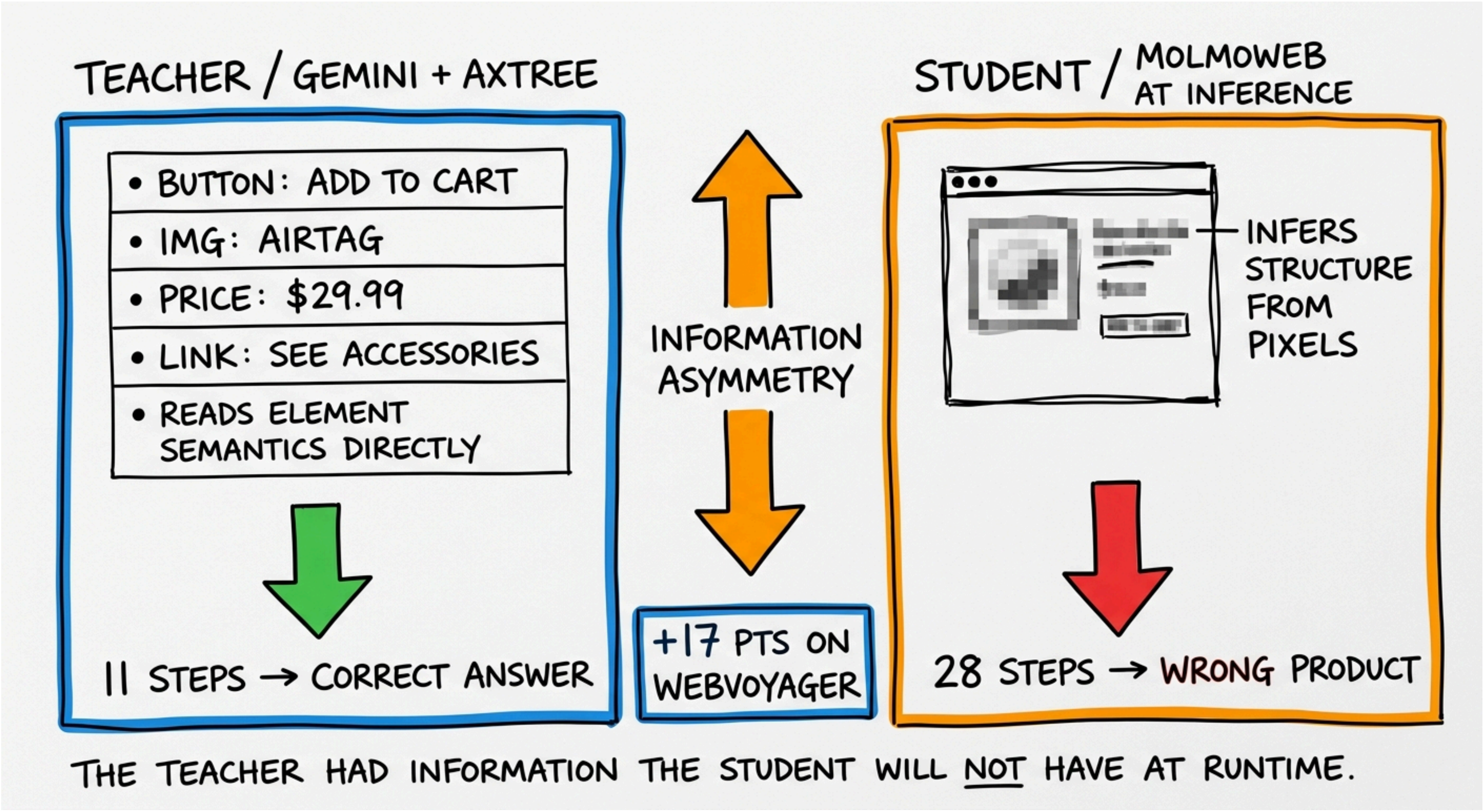
Direct path to the correct answer

What MolmoWeb actually is

- The **Molmo 2** VLM (SigLIP2 vision + Qwen3 backbone) — not a new architecture
- Trained by **supervised fine-tuning alone** — no RLHF, DPO, or self-distillation
- Sees a **screenshot, URL, and the last 10 actions**; emits a ReAct thought + action
- Click coordinates are floats normalized to **[0, 100]** — Molmo's pointing convention
- **Pure pixels at inference** — no DOM, no accessibility tree, no element overlay

The teacher sees what the student won't

"Synthetic" doesn't mean an AI looked at screenshots. A text-only teacher read the page's structure — information the trained model never gets at runtime.



Build better teachers, not bigger crawls

Most open agents distill from GPT-4o — TOS- and reproducibility-fragile. MolmoWeb builds its own teachers from **text-only Gemini reading the accessibility tree**.

01

Planner

A Gemini planner decomposes the task into a sequence of steps.

02

Operator

A Gemini agent reads the AxTree and acts on element semantics — not pixels.

03

Verifier

A GPT-4o screenshot check plus a WebVoyager judge filters trajectories before training.

Multi-agent beats single-agent 78.5 vs 74.4 — the orchestration earns its complexity.

At matched scale, synthetic wins

BENCHMARK (2.7K RUNS)	SYNTHETIC	HUMAN	GAP
WebVoyager	53.0	35.4	+17.6
Online-Mind2Web	16.8	9.0	+7.8
DeepShop	24.4	19.8	+4.6

The gap doesn't narrow at scale. Adding 28K human trajectories on top of the full synthetic set is a wash — **human demos add no meaningful signal**.

Curation beats volume

A second MolmoWeb ablation lands harder than the headline: **10% of the training mixture reaches 85–90% of full-data performance**. The mixture is data-redundant by design.

10%

of the mixture → 85–90% of full-data performance

90%

of trajectories add only marginal signal

?

how to find the right 10% before training — unsolved

Volume isn't the variable that matters. Curation is.

Consensus aggregates confusion

Governance assumes human labels are "noisy but unbiased" — random error cancels.
Web navigation breaks that assumption: the noise is **structural**.

- Annotators share the same confusion → **correlated** exploratory paths
- Inter-annotator agreement stays **high** — everyone is confused the same way
- Training loss drops **normally** — the model learns to imitate the hedging
- Failure surfaces only at evaluation, where the model hedges when it should act

COMPREHENSION CHECK · PART 1

Training-data governance assumes human labels are “noisy but unbiased.” Why does web-navigation annotation violate this?

- **A** Crowdwork selects for speed over accuracy, a throughput bias
- **B** The noise is structural — shared confusion makes consensus aggregate correlated error
- **C** Per-task pay creates a financial-incentive bias on hard tasks
- **D** Interfaces drift during collection, causing temporal label drift

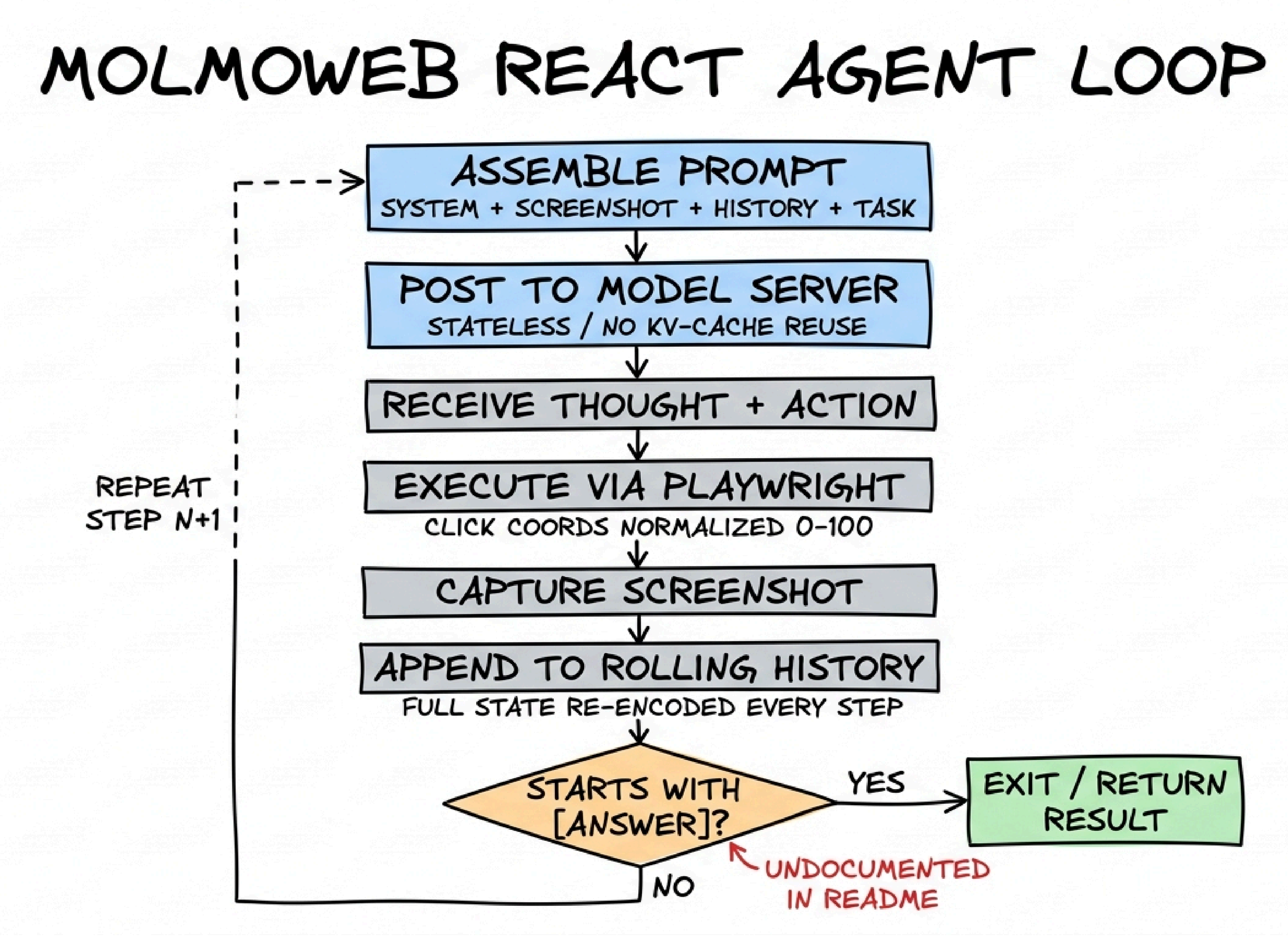
PART 2 · INSIDE THE HARNESS · MAY 4, 2026

02

Inside the MolmoWeb Harness

Six weeks after launch, Ai2 shipped the training code, benchmark runner, and annotation tool. Now that everything is visible, we can read what “complete” actually means.

What "complete" reveals: the ReAct loop



The repo went complete

- `training/`, `benchmarks/`, and `annotation/` are now populated
- Weights + data + training code + eval suite + annotation tool — all **Apache 2.0 / ODC-BY**
- For GUI agents, **nothing else is this open** at this performance level
- But: complete on paper, **operationally rough** in practice

The defaults don't match the paper

RUN_TRAIN.SH	SCRIPT	PAPER	GAP
CHECKPOINT	4B	8B	**size**
NUM_GPUS	8	64	**8×**
GLOBAL_BATCH	64	128	**2×**
DURATION	500	50,000	**1%**

A naive `bash run_train.sh` trains ~1% of paper with no warning in the README

The headline number isn't turnkey

94.7%

pass@4 on WebVoyager — the figure most often cited

- pass@4 means **four parallel rollouts per task**, not one
- The parallel-rollout orchestration **isn't in the repo** — you build it
- Stacked on the 1%-budget training default, independent validation is real work
- Open weights ≠ a turnkey reproduction of the headline

Three concerns that survived the code drop

01

LLM-judge loop

GPT-4o filters which synthetic trajectories the model learns from — and is also the WebVoyager benchmark judge.

02

ScreenSpot dual-use

Listed as both a training input and the headline grounding benchmark, with no documented split-separation audit.

03

No deduplication

Training domains and eval benchmarks share the same pool, with no programmatic success checker independent of the judge.

COMPREHENSION CHECK · PART 2

GPT-4o filters the synthetic training trajectories AND judges the WebVoyager benchmark. Why does the post flag this loop?

A Reusing one model for data and eval is banned by ML-venue guidelines

B The same model gates training and scores success — filter and eval share an inductive bias

C GPT-4o approves its own style by a fixed, correctable percentage

D Ablations prove it inflates scores by a specific subtractable amount

No allowlist. No injection defense.

- No URL allowlist, no blocklist, **no prompt-injection mitigation** for on-screen text
- Any page the agent visits can display text attempting to redirect it
- The author's documented mitigation: *"don't ask the model to do sensitive things"*

A GUI agent inside a bank is by definition asked to do sensitive things — that is the entire point of deploying it.

The silent-failure pattern

- The script runs and produces a model — at 1% compute, **no warning**
- The general pattern: **behavioral assumptions never verified** against actual configuration
- **SR 26-2** (April 2026) carves GenAI and agentic AI out of scope
- This is exactly what **effective challenge** is designed to catch — and rarely does under pressure

03

PART 3 · THE VERIFICATION GAP · MAY 5, 2026

Did It Actually Work?

The canonical verifiers behind GUI agent benchmarks have false-positive rates above 40% — not from weak models, but from how they're architecturally designed.

PART 3 · THE PROBLEM

The scoreboard has been wrong

On WebVoyager with the Fara-7B agent, the native verifier and the Universal Verifier disagree by a wide margin in one consistent direction.

75%

Native verifier calls these tasks successful

38%

Universal Verifier calls successful

60%

False-positive rate — successes that were failures

Four principles, one root

1 Rubric before output

Generate criteria from the task alone — before seeing what the agent did. No post-hoc rationalization.

2 Process vs outcome

Score how faithfully each sub-goal ran, separately from whether the user's goal was achieved.

3 Control the uncontrollable

CAPTCHA walls and outages shouldn't penalize the process score; reasoning errors should.

4 Relevance matrix

Score each screenshot against every criterion, so a state change at step 37 isn't lost in bulk context.

Rubric quality drives half the gains

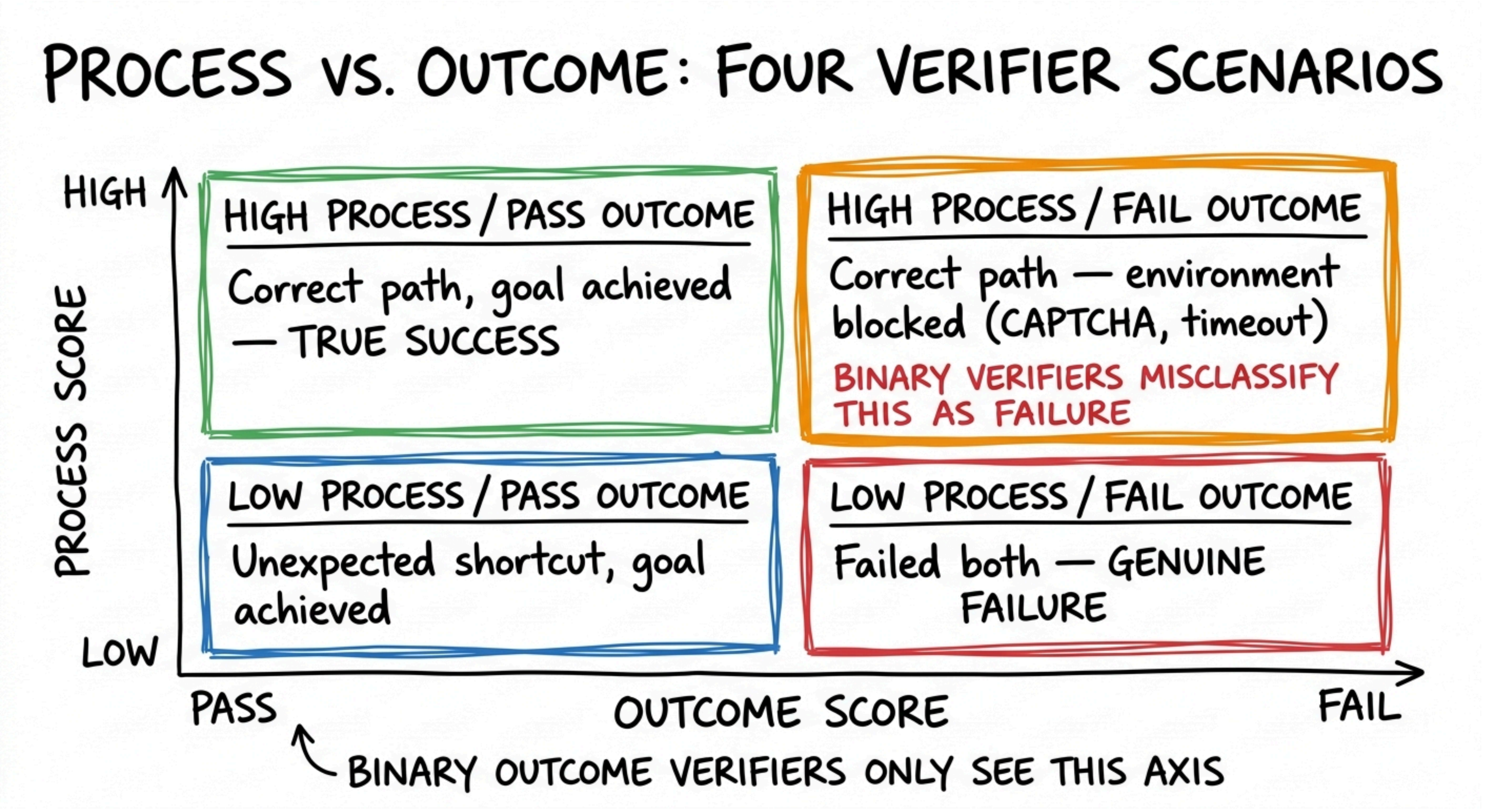
~50%

of the Cohen's κ gains (across 96 experiments) come from rubric quality alone

- Generate the rubric from the task alone — **before** seeing what the agent did
- Score separately: a same-call rubric rationalizes the agent's actual behavior
- Two-pass hallucination check — score with and without screenshot evidence
- MRM parallel: separating rubric author from scorer **is** independent validation

Process ≠ outcome

For a bank, "the agent followed the compliance workflow" and "the output was correct per policy" are **different audit questions**.



70% of expert, in 5% of the time

1 day

Auto-research agent (Claude Opus 4.6 + Claude Code) vs the expert's three weeks

~70%

Of expert-level Cohen's κ, from blank prompts

But its edits were "conservative and incremental" — it couldn't encode the evaluative judgment behind the **structural** changes that drove the expert's step-function gains.

Three questions left open

01

Transfer

Do the principles hold in closed enterprise environments? Likely — but rubric generation gets substantially harder.

02

Goodhart

Use the verifier as an RL reward and agents learn to satisfy its rubric — which may diverge from real user value.

03

Regulation

SR 26-2 excludes agentic AI; no framework specifies what “adequate verification” means in that context.

Systematic error at every layer

PART 1	**Training signal**	Human annotations carry structured noise the model learns to imitate.
PART 2	**Implementation**	Defaults, dataset dual-use, and a judge loop mean papers may not reproduce.
PART 3	**Measurement**	A 60% false-positive rate means the benchmarks overcounted success for years.

Each layer erred toward overstated capability. The field spent two years building better agents; maybe it should have been building better evaluations.